

Histogram Equalization and Lighting Correction Report

Intensity Transformation

The implementation processes the 24-bit 'moon.bmp' by converting it to a grayscale matrix. Histogram equalization starts by computing a 256-bin histogram to record the frequency of each intensity value. The script derives a probability mass function from these counts and computes the cumulative distribution function. This cumulative distribution is scaled to the 0 to 255 range to create a mapping table. Applying this table to the image redistributes the pixel values to produce a flatter histogram. This process increases the local contrast of the lunar surface features.

Surface Fitting and Pseudo-Inverse

Lighting correction follows the least squares method discussed in class. The code defines a coordinate matrix A whose rows contain the spatial positions of pixels. For the linear method, these rows consist of the horizontal position, vertical position, and a constant. The quadratic method expands this to include squared terms and the product of coordinates. The script solves the system $Ax = t$ by applying the normal equations $(A^T A)x = A^T t$. This calculates the coefficients for the lighting surface using the pseudo-inverse logic. Subtracting this fitted surface from the original data removes global lighting gradients.

Scaled and Truncated Methods

The script produces two distinct outputs for both linear and quadratic corrections. The scaled version subtracts the surface and shifts the minimum value to zero. It then multiplies the result by a constant to fit the maximum value to 255. This preserves all data without saturation. The truncated version subtracts the surface and adds the global mean intensity of the original image back to the values. Any results falling outside the 0 to 255 range are clipped at the boundaries. This maintains a brightness level similar to the input but can lead to a loss of detail in extreme highlights or shadows.

Implementation Results

The quadratic model captures curved lighting patterns that the linear plane cannot approximate. Using the Pillow library for initial conversion ensures the grayscale values remain consistent with standard luminance formulas. Matplotlib is used to export the final PNG files into the results folder. The results show that the quadratic scaled version provides the highest level of detail without the blackouts found in the truncated versions. All calculations rely on NumPy to maintain numerical stability during the matrix inversion process.